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Genome-wide identification of wheat Ubiquitin-conjugating enzyme gene family and its role in wheat powdery mildew disease resistance

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Abstract

Ubiquitin-conjugating enzyme (UBC) E2 is a critical component of the ubiquitination process. It interacts with the RING domain of E3 ligases through its UBC domain, enables specific identification of substrate proteins, thereby plays a pivotal role in plant growth and development. In this study, the number, gene structure, and evolutionary relationships of *UBC* genes in wheat and *Arabidopsis thaliana* were analyzed at a whole-genome level. Furthermore, the roles of *TaUBC* genes in wheat resistance to powdery mildew were investigated. A total of 180 *UBC* genes were identified in wheat and categorized into 15 Classes through phylogenetic analysis. Quantitative real-time PCR analysis revealed significant expression changes of *TaUBC1A-5*, *TaUBC1A-6*, *TaUBC3A-5*, *TaUBC3B-14*, *TaUBC5A-2*, *TaUBC6A-2*, and *TaUBC7A-2* after the powdery mildew infection, compared to the 0-h time point. Virus-induced gene silencing experiments demonstrated that *TaUBC3A-5* positively regulates wheat resistance to powdery mildew, whereas *TaUBC3B-14* acts as a negative regulator. These findings provide valuable insights into the function of the *UBC* gene family in wheat and offer support for development of new disease-resistant wheat germplasm using *UBC* genes.

Keywords Wheat, *UBC* gene family, Powdery mildew, Gene expression, Virus-induced gene silencing

Background

The ubiquitin (Ub) /proteasome pathway represents the primary protein degradation mechanism in higher plants. Through a three-step enzymatic reaction, ubiquitin is covalently attached to the target protein substrate, marking it for degradation by the 26S proteasome [21]. The process of protein ubiquitination is a critical regulatory mechanism for protein turnover in plants. It primarily involves the coordinated action of E1 (Ub-activating enzyme, UBA), E2 (Ub-conjugating enzyme, UBC), E3 (Ub ligase), the 26S proteasome, and ubiquitin itself [12, 36, 58]. This system plays a key role in various aspects

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of plant growth and development, including cell cycle regulation, programmed cell death, plant morphogenesis, self-incompatibility, as well as the regulation of light, sugar, hormone, and nutrient metabolism and signal transduction pathways [28, 38, 39, 41, 51, 57, 63].

E2 is a key enzyme in the ubiquitination process, characterized by a conserved UBC catalytic domain consisting of 140 to 220 amino acids. This domain interacts with the RING domain of E3 ligases to facilitate the specific recognition of substrate proteins, which subsequently influences plant morphogenesis [40]. E2 enzymes that lack cysteine active sites within the UBC domain are referred as ubiquitin-conjugating E2 enzyme variants (UEVs), and these UEV proteins lack enzymatic catalytic activity, however they may fulfill other critical biological functions in plants [4].

Ubiquitin-binding enzymes related to plant development and DNA repair process. In *Arabidopsis*, *AtUBC2* inhibits the flowering process [57]. The knockout mutant of the *AtUBC22* gene produces shorter and fewer seeds in siliques [48]. In tomato, the ripening inhibitor (RIN) protein directly binds to the E2 promoter region to regulate fruit ripening. In the nucleic acid repair pathway, the UBC35/UBC36 and Mms2p/Uev1a proteins form complexes that initiate signal transduction in nucleic acid repair processes to facilitate the restoration of damaged nucleic acids [14]. The RNF8-UBC13 complex independently mediates the ubiquitination of substrate proteins and promotes the accumulation of the DNA damage regulator 53BP1 protein [18].

The E2 gene family plays a vital role in responding to abiotic stress factors. *AtUBC32* is involved in brassinosteroid-mediated salt stress tolerance and mutation of *AtUBC32* enhances tolerance to tunicamycin-induced endoplasmic reticulum stress [9, 46]. Overexpression of the tobacco gene *NtUBC1* enhances resistance to cadmium stress [2]. The wild rice gene *OgUBC1* exhibits enhanced expression in leaves when exposed to exogenous salicylic acid and UV-B irradiation [19]. Overexpression of the *VrUBC1* gene in mung bean promotes abscisic acid- and salt-mediated stomatal closure, thereby improving drought resistance [8]. *GmUBC2* is widely expressed in various tissues, and its overexpression in *Arabidopsis* enhances tolerance to both salt and drought stresses [63].

The E2 gene family also contributes to enhancing plant immunity against pathogens in different plant species [37]. In *Arabidopsis*, *AtUBC13* regulates the programmed cell death in response to low temperature and pathogen, and mutation of *AtUBC13* shows compromised resistance to avirulent bacterial [47]. On the contrary, the ortholog protein OsUBC13 in rice negatively regulates rice resistance to *Magnaporthe oryzae* and *Xanthomonas oryzae* pv *oryzae* [29]. Reducing the expression

levels of *FNi3*, a *UBC13* ortholog in tomato, affects *Fen*-mediated immunity and disrupts cell death triggered by other resistance proteins [37]. In tobacco, Group III E2s are vital for the production of immune-related reactive oxygen species and the suppression of immune-related programmed cell death [62]. Silencing the *UBC2* gene in tobacco enhances the resistance of transgenic plants to tobacco mosaic virus [45]. In recent years, some studies reported that the *UBC* genes were involved immunity in rice. The *OsUBC26* showed resistance to *Magnaporthe oryzae* together with E3 ligases through regulating the expression of *WRKY45* [31]. Overexpression of *OsUBC45* showed enhanced broad-spectrum resistance to bacterial leaf blight and blast [50]. In wheat, *TaU4* was reported to negatively regulate wheat resistance to *Septoria* leaf blotch [34].

As a major source of global food, wheat faces numerous biotic stresses during its growth, such as rust, Fusarium head blight (FHB) and powdery mildew, which significantly affect wheat yield, quality, and even human and animal health [15, 16, 27, 42, 53, 56]. Identifying resistance genes to those diseases in wheat and growing resistant varieties with those genes represents one of the most effective and environmental safe strategies for crop protection [3, 16, 25]. Although previous studies have highlighted the role of *UBC* genes in plant growth, development, and responses to abiotic stresses, their involvement in biotic stresses remains to be explored. In this study, the number, evolutionary relationships, and physicochemical properties of the wheat *UBC* gene family, along with their roles in resistance to powdery mildew and FHB, were analyzed. The findings provide valuable information for the functional research of the wheat *UBC* genes and provide support for the development of disease-resistant wheat germplasm using *UBC* genes.

Materials and methods

Plant materials and treatment

Two common wheat varieties Sumai 3, which carries the *Fhb1* gene [27, 43] and Bainong 207, which is susceptible to mixed *Blumeria graminis* f. sp. *tritici* (*Bgt*), were used for gene expression analyses. The common wheat variety Bainong AK58 (AK58) was used for the barley stripe mosaic virus-induced gene silencing (BSMV-VIGS) assay, growing in a controlled condition with 14 h of light at 16 °C, 10 h of darkness at 12 °C, and 70% relative humidity. Bainong 207 and AK58 were developed and maintained by the Agriculture College of Henan Institute of Science and Technology, Xinxiang, Henan, China, and Sumai 3 was collected and maintained by the same institution. The mixed races of *Bgt* were collected from fields in Xinxiang, Henan, China, and maintained on seedlings of the susceptible wheat variety Sumai 3 under the conditions of 22 °C/18 °C, a

14-h light/10-h darkness cycle, and 70% relative humidity. Bainong 207 was grown under the same conditions as Sumai 3 but without exposure to *Bgt*. Seedlings of Bainong 207 were inoculated with *Bgt* at the two-leaf stage. Field isolates of *Fusarium graminearum* were isolated from a wheat field in Henan, China. Sumai 3 was grown in a field in Xinxiang and spikelets of Sumai 3 were inoculated with *F. graminearum* using the single-floret inoculation method at the early flowering stage, as previously described [17].

Total RNAs were extracted from the leaf tissues of Bainong 207 inoculated with *Bgt* at 0, 3, 6, 12, 24, 48 and 72 h post-inoculation (hpi) and the spikelets from different individuals of Sumai 3 inoculated with *F. graminearum* at 0, 3, 6, 12, 24, 48 and 72 hpi using Trizol reagent (Vazyme, Nanjing, China) following the manufacturer's protocol. The different time points after pathogen inoculation select for the gene expression analysis was based on the developmental process of pathogens and the plant's immunity system [61].

Quantitative RT-PCR analysis of *TaUBC*

RNA samples were reverse-transcribed using the HiScript® III RT SuperMix for qPCR kit (Vazyme, Nanjing, China). Relative expression levels of the target genes were quantified via qRT-PCR using a SYBR Green detection kit and the AceQ qPCR SYBR Green Master Mix (Vazyme, Nanjing, China) on an LC 480 II system (Roche, Germany). The wheat gene *TaTubulin* served as an internal control. The qRT-PCR started with an initial denaturation at 95 °C for 5 min, followed by 40 cycles of denaturation at 95 °C for 10 s and extension at 60 °C for 20 s. Relative gene expression levels were calculated using the comparative $2^{-\Delta\Delta CT}$ method. All primers used in this study (Table S1) were synthesized by Gene Create (Wuhan, China). The qRT-PCR data is presented in Table S2.

BSMV-VIGS

According to the gene evolutionary relationship and the expression pattern induced by *Bgt*, several target genes were selected to verify their roles in wheat powdery mildew resistance using BSMV-VIGS. To silence the target *TaUBC* genes, fragments of *TaUBC3A-5* (*TaUBC3B-5* and *TaUBC3D-4*) with 199 bp and *TaUBC3B-14* (*TaUBC3A-14* and *TaUBC3D-13*), with 205 bp were amplified using corresponding primer pairs (Table S1) and reversely inserted into the γ -strain of BSMV to produce the vectors BSMV: *TaUBC3A-5* and BSMV: *TaUBC3B-14*, respectively. The second fully unfolded leaves of AK58 were infiltrated with the transcribed virus carrying BSMV: TaPDS using BSMV: γ -infected leaves as controls. The infiltrated plants grew at 23 °C, under a 14 h light/10 h dark cycle at 70% relative

humidity. When the fourth leaves showed symptoms of viral infection, the leaves were collected and inoculated with mixed *Bgt* to evaluate disease resistance by culturing them on a 6BA-plate with mixed powdery mildew spores under conditions of 14 h light at 22 °C and 10 h/darkness 18 °C for 6 days. Target gene silencing efficiency was evaluated by qRT-PCR using the corresponding primer pairs (Table S1).

Genome-wide identification of *UBC* family members

The genome sequence (*Triticum aestivum*.IWGSC.pep.all.fa and *Triticum aestivum*.IWGSC.50.gff3) of wheat (*Triticum aestivum*, 'Chinese Spring') was downloaded from the International Wheat Genome Sequencing Consortium (IWGSC) and accessed via Ensembl Plants (https://ftp.ensemblgenomes.ebi.ac.uk/pub/plants/release-60/fastq/triticum_aestivum/). Additionally, the genome sequence (*Arabidopsis thaliana*.TAIR10.pep.all.fa and *Arabidopsis thaliana*.TAIR10.50.gff3) of *Arabidopsis thaliana* was obtained from the same source (https://ftp.ensemblgenomes.ebi.ac.uk/pub/plants/release-60/fastq/arabidopsis_thaliana/). The typical UBC domain (PF00179) was downloaded from the Pfam database served as the search model. A new Hidden Markov Model (HMM) was constructed to ensure the accuracy of search results as described by Xu et al. [55]. Using the newly established HMM as the search model, the HMMERv3 suite was employed with the hmm-build function (E-value $< 1 \times 10^{-20}$) to identify UBC across the complete genomes of wheat and *Arabidopsis*. Candidate sequences were subsequently validated using SMART [26] and NCBI-CDD [32]. Only sequences containing complete UBC conserved domains were retained for further analysis (Table S3).

Phylogenetic analysis, gene structure, and conserved motif analysis

Using the maximum likelihood method in MEGA-X software [24], a phylogenetic tree for the *UBC* family members of wheat and *Arabidopsis* was constructed with 1000 bootstrap replicates. The phylogenetic tree was visualized using EvolView [13]. Exon–intron structure analysis was performed using the wheat GFF3 file, and motif analysis was conducted with the MEME program (<http://meme.uite.org/tools/meme>). The maximum number of motifs was set to 20, with an optimal width for each motif ranging from 6 to 50 residues [55]. The structural features motif compositions of *UBC* genes were visualized using TBtools [7].

Gene replication

To identify gene duplication events, the Multiple Collinearity Scan toolkit (MCScanX) was utilized [49]. Gene duplication events, syntenic relationships among gene pairs, and chromosome localization of the analyzed

genes were visualized using ShinyCircos software (<http://shinycircos.ncpgr.cn>) [59]. Ka/Ks values of tandem repeat genes were calculated using TBtools [7].

Cis-acting element and transcriptomic analysis of wheat *UBC* members

The 2,000 bp sequence upstream of the *UBC* was submitted to the Plant CARE (<http://bioinformatics.psb.ugent.be/webtools/plantcare/html/>) for predicting cis-acting element, and the results were visualized using TBtools [7]. RNA-seq data (Table S3) was obtained from WheatOmics (<http://wheatomics.sdau.edu.cn/expression/wheat.html>) [33]. Data of powdery mildew responses was collected from N9134 (a powdery mildew resistant wheat variety) [60]. The relative expression levels of each *TaUBC* gene under powdery mildew stress was visualized as a heat map generated using TBtools [7].

Protein structure and active sites

Protein tertiary structure predictions were conducted using SWISS-MODEL (<https://swissmodel.expasy.org/>). The template of *TaUBC3A-5* is A0A1D6RUR5.1.A, and the GMQE score is 0.92. The template of *TaUBC3B-14* is Q6L508.1.A, and the GMQE score is 0.94 (Supplementary-PDB). Active protein sites were predicted with SPPIDER (<http://sppider.cchmc.org/>) (Table S4), and the protein structures were visualized using PyMOL [35].

Results

Identification, phylogenetic and conserved motif analysis of wheat *UBC* family members

A total of 180 and 48 *UBC* genes were identified in the genomes of wheat and *Arabidopsis*, respectively. The *UBC* family members were named based on their chromosomal locations, and a phylogenetic tree was constructed for the *UBC* family members of wheat and *Arabidopsis* (Fig. 1). The phylogenetic analysis classified the 228 *UBC* members into 15 Classes, designated as Class 1 to Class 15. Class 11 contained the most members (54), while Classes 8 and 13 each had the fewest (only five each). The presence of both wheat and *Arabidopsis* *UBC* genes within each Class suggests that the differentiation of the *UBC* gene family occurred prior to the divergence of monocots and dicots.

The structural characteristics of wheat *UBC* family members are presented in Fig. 2 and Figure S1. In the Classes 1, 3, 5, 7, 8, 9, and 10, all wheat *UBC* members contain Motifs 1 and 2 (Fig. 2). The Classes 3 and 9 also include Motif 3. All Classes contain Motif 4 except for the Class 9. Motif 5 is only present in Classes 1, 3, 5, and some members of Class 10. These findings indicate that the motif compositions of wheat *UBC* gene family members within the same Class is highly similar, whereas the motif compositions across different Classes may vary

(Fig. 2 and Figure S1). For gene structure, the members of Class 1 and 9 each contain 4–5 exons, while members of Classes 3, 5, 7, and 10 typically contain 5–6 exons (Fig. 2C). In the Class 8, all members, except *TaUBC3A-9* with three exons, contain two exons. The exon–intron structures of wheat *UBC* genes within the same subfamily are more conserved, further supporting the conservation of the *UBC* gene family.

Chromosome locations, duplications and Ka/Ks analysis of *UBCs*

A total of 179 *UBC* genes were identified to be unevenly distributed across the 21 wheat chromosomes (Fig. 3), while the chromosomal location of one *UBC* gene remains undetermined. Common wheat, being a hexaploid species with A, B, and D subgenomes, harbored 58, 59, and 62 *UBC* genes in the A, B, and D subgenomes, respectively. The *UBC* genes were predominantly located on homoeologous groups 1, 3, and 5, with 31, 46, and 30 genes, respectively. Homoeologous group 7 exhibited the lowest number of *UBC* genes, with only 14 identified genes. A total of 111 segmental gene pairs and 7 tandem repeats were identified among the wheat *UBC* genes (Fig. 3). The Ka/Ks ratios of tandem repeat genes were all less than one, indicating that they have undergone purifying selection and have stabilized (Table 1). Most segmental gene pairs were located among the A, B, and D subgenomes, reflecting segmental duplications among the three subgenomes. Eight transchromosomal segmental duplication events were identified, such as *TaUBC4A-6* and *TaUBC5B-11*, and *TaUBC5A-8* between chromosomes 4A and 5B. Other segmental duplications included *TaUBC1D-4* and *TaUBC3A-5*, *TaUBC1D-7* and *TaUBC4B-5*, as well as *TaUBC1D-8* and *TaUBC3A-14*. Additional examples included *TaUBC1D-9* and *TaUBC3A-13*, *TaUBC1D-10* and *TaUBC3A-11*, and *TaUBC1D-12* and *TaUBC3A-12*. Additionally, seven tandem repeats were identified, including gene pairs *TaUBC3B-1* and *TaUBC3B-2*, *TaUBC3B-7* and *TaUBC3B-8*, *TaUBC3D-1* and *TaUBC3D-2*, as well as *TaUBC3D-6* and *TaUBC3D-7*. Other tandem repeats included *TaUBC5D-7* and *TaUBC5D-8*, *TaUBC6A-5* and *TaUBC6A-6*, and *TaUBC6B-4* and *TaUBC6B-5*, located on chromosomes 3B, 3D, 5D, 6A, and 6B, respectively. Overall, most of the *UBC* genes studied originated from their ancestral species, while a smaller proportion underwent exchanges or duplications during evolution, reflecting the complex evolutionary history of the *UBC* gene family in wheat.

Cis-acting components, heat map and qRT-PCR of *UBC* family members

The cis-regulatory elements of *UBC* are mainly divided into three categories: hormone-responsive elements, stress-responsive elements and plant growth and

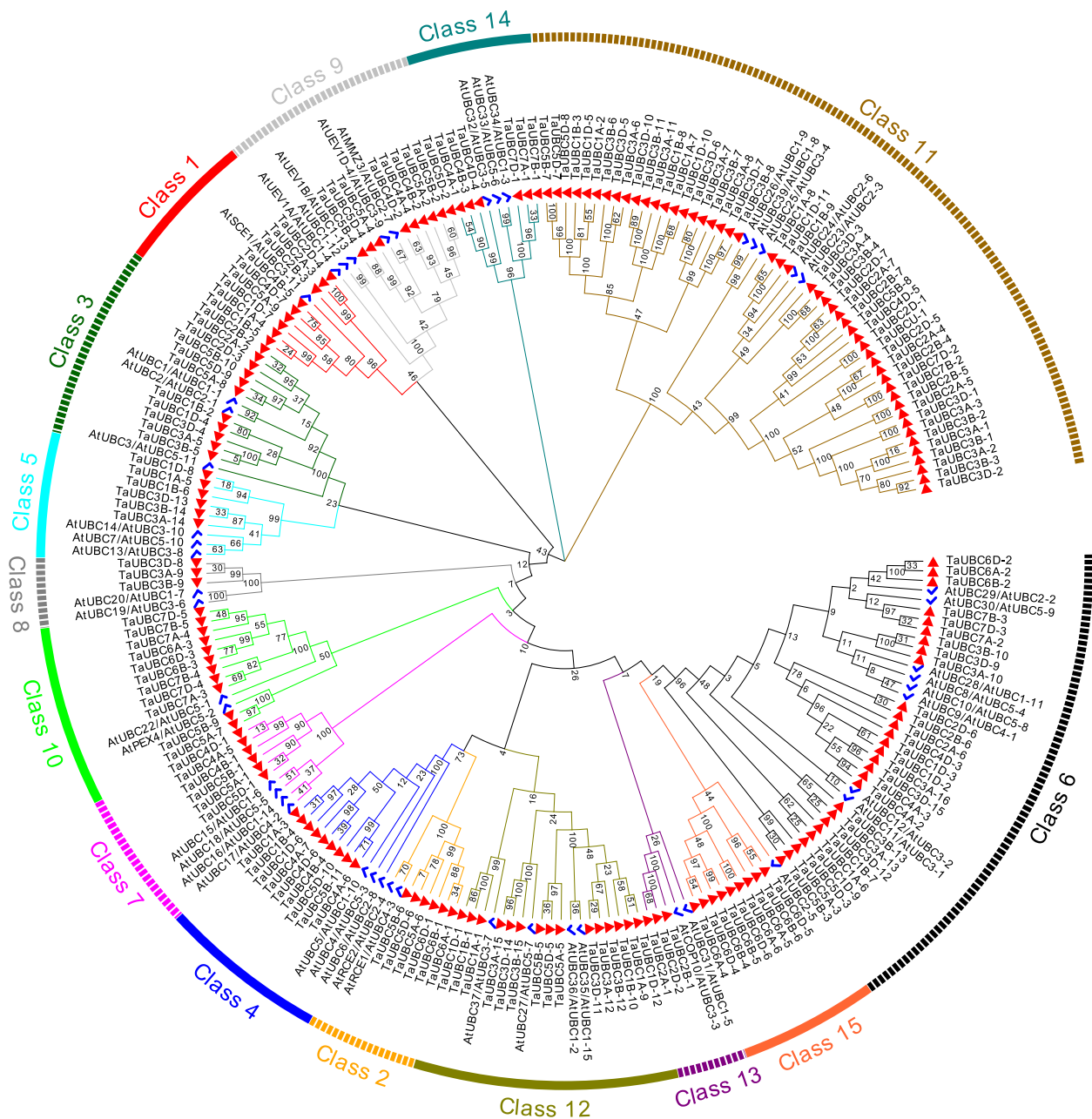


Fig. 1 Phylogenetic Analysis of UBC Protein in *Triticum aestivum* and *Arabidopsis thaliana*. Wheat UBC proteins are indicated by red triangle, Arabidopsis UBC proteins are indicated by blue check

development elements (Figure S2). The expression levels of wheat *UBC* genes exhibited diverse trends under powdery mildew stress (Fig. 4). The expression levels of genes in Classes 7, 8, and 10 showed minimal variation, whereas the expression levels of genes in Classes 1, 3, 5, 9, 12, and 13 initially increased and then decreased as the duration of stress extended. Compared with the non-inoculated plants, the expression level of *TaUBC5A-9* in Class 1 increased by 1.57-fold at 48 hpi. Similarly, in Class 3, the expression levels of *TaUBC3A-5* and *TaUBC5A-8* were elevated by 1.40-fold and 1.87-fold, respectively,

at 48 hpi. The expression of *TaUBC1A-5* in Class 5 and *TaUBC4A-4* in Class 9 increased significantly, by 2.14-fold and 2.56-fold, respectively, at 48 hpi.

To further investigate the role of *TaUBC* genes in response to powdery mildew and *F. graminearum* stresses, qRT-PCR analysis was conducted. Under *F. graminearum* stress (Fig. 5), compared with 0 hpi, the expression levels of *TaUBC1A-5*, *TaUBC1A-6*, *TaUBC2A-6*, *TaUBC2D-2*, *TaUBC3A-5*, *TaUBC3A-7*, *TaUBC3A-13*, *TaUBC3B-14*, *TaUBC4D-3*, and *TaUBC5A-2* increased significantly as the stress duration prolonged, with fold changes ranging

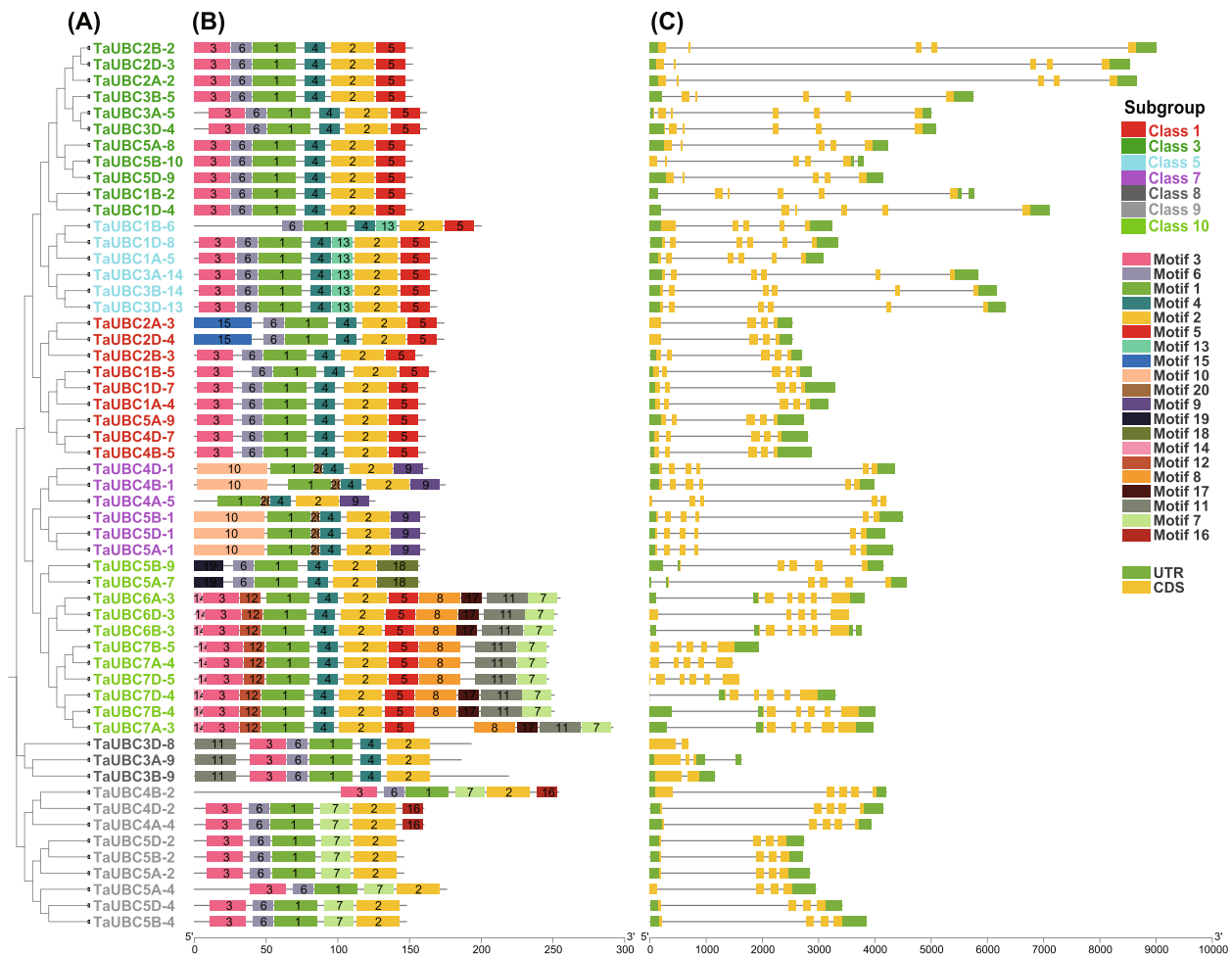


Fig. 2 The gene structure and conserved domain organization of UBC genes in *Triticum aestivum*. **a** The evolutionary tree was built by NJ method. **b** Motif distribution of UBC proteins. **c** Exon–intron structure of UBCs

from 1 to 2.5 folds. Conversely, *TaUBC3A-10*, *TaUBC6A-2*, *TaUBC6A-3*, and *TaUBC7A-2* showed an initial decrease followed by an increase in expression levels, also the difference reached statistical significance.

Under *Bgt* stress (Fig. 6), the expression levels of *TaUBC* genes showed a pattern of first increasing and then decreasing as the duration of stress increased, with statistically significant differences. Except for *TaUBC6A-3*, the expression levels of all *TaUBC* genes peaked at either 3 hpi or 6 hpi. Notably, some genes were strongly upregulated, including *TaUBC1A-6*, *TaUBC3A-5*, *TaUBC3B-14*, *TaUBC5A-2*, *TaUBC6A-2*, and *TaUBC7A-2* with fold changes of 7.85-, 7.13-, 8.08-, 6.01-, 5.74-, and 17.98-fold, respectively, at 3 hpi. These results suggest that certain *TaUBC* genes may be more actively involved in the response to *Bgt* than to *F. graminearum*.

Silencing *TaUBCs* changed the resistance of AK58 to powdery mildew

Since the relative expression levels of *TaUBC3A-5* (*TaUBC3B-5* and *TaUBC3D-4*) and *TaUBC3B-14*

(*TaUBC3A-14* and *TaUBC3D-13*) were induced by *Bgt* (Figs. 6E and 6I), they were selected to investigate their potential roles in powdery mildew resistance in AK58 using the VIGS. The newly constructed vectors BSMV: *TaUBC3A-5* and BSMV: *TaUBC3B-14*, carrying 199 bp and 205 bp fragments, respectively, were used to induce silencing of the target genes. The fourth leaves of plants infected with BSMV: *TaUBC3A-5* and BSMV: *TaUBC3B-14* were used as the RNA source for qRT-PCR analysis and resistance evaluation. Plants infected with BSMV:γ served as controls. qRT-PCR results revealed significantly reduced expression of *TaUBC3A-5* and *TaUBC3B-14* in plants infected with either BSMV: *TaUBC3A-5* or BSMV: *TaUBC3B-14*, compared with plants infected with BSMV:γ (Fig. 7 A1 and B1). Six days after *Bgt* inoculation, phenotypic analysis showed that leaves infected with BSMV: *TaUBC3A-5* were more susceptible to *Bgt* than those infected with BSMV:γ (Fig. 7A2). Conversely, leaves infected with BSMV: *TaUBC3B-14* exhibited reduced susceptibility to *Bgt* compared with BSMV:γ-infected plants (Fig. 7B2). These results suggest that silencing *TaUBC3A-5* increases susceptibility

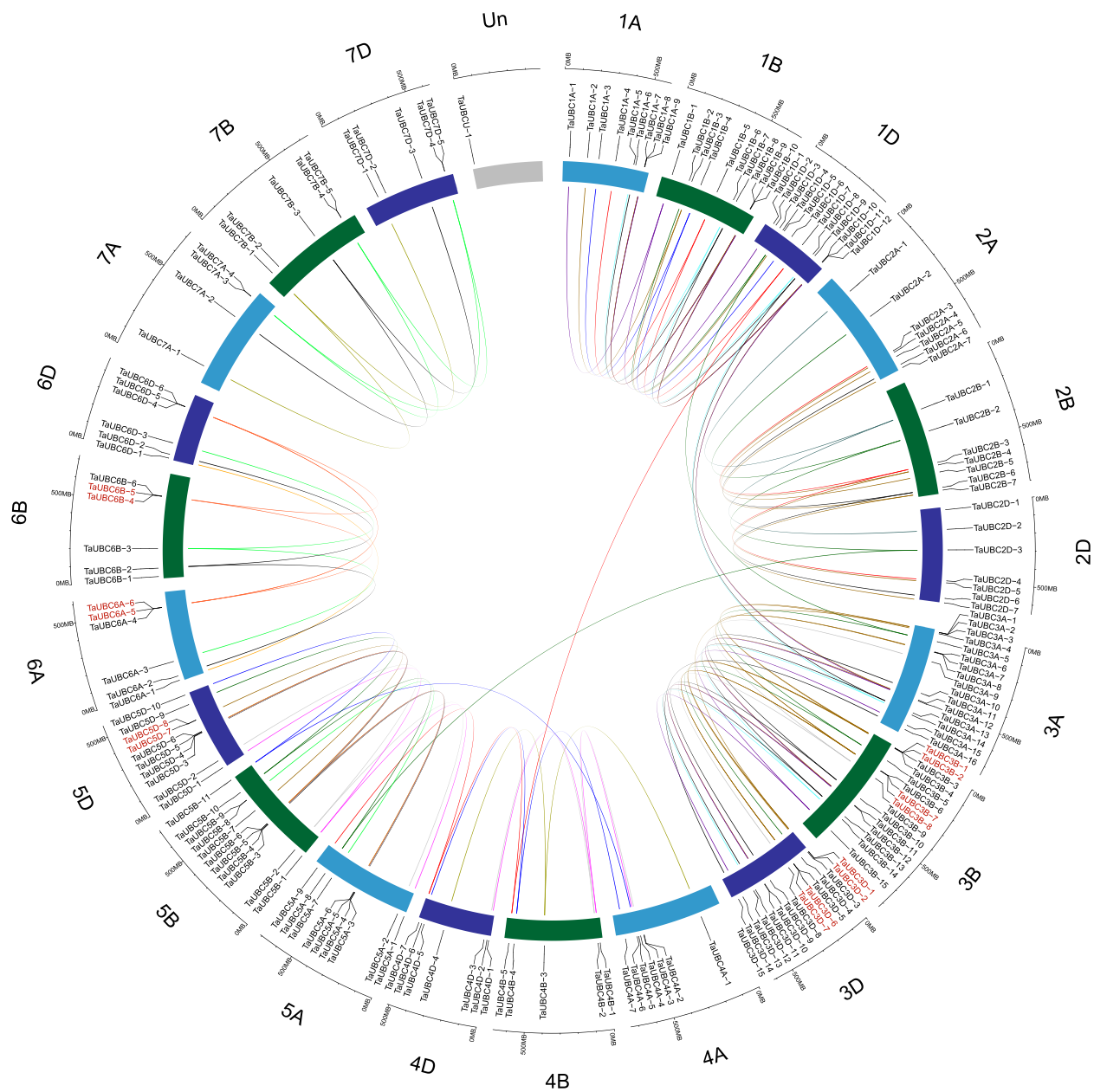


Fig. 3 Location, homologous gene pairs and tandem duplication of *UBC* genes in *T. aestivum*. The adjacent red *UBCs* represent tandem repeat genes

Table 1 The Ka/Ks ratios for tandem duplication

Tandem duplication	Ka	Ks	Ka/Ks	Purifying selection
TaUBC3B-1/TaUBC3B-2	0.0199	0.0551	0.3619	Yes
TaUBC3B-7/TaUBC3B-8	0.1157	0.3479	0.3326	Yes
TaUBC3D-1/TaUBC3D-2	0.1050	0.3364	0.3121	Yes
TaUBC3D-6/TaUBC3D-7	0.1254	0.3832	0.3272	Yes
TaUBC5D-7/TaUBC5D-8	0.1373	0.2788	0.4924	Yes
TaUBC6A-5/TaUBC6A-6	0.1248	0.2326	0.5367	Yes
TaUBC6B-4/TaUBC6B-5	0.0045	0.0328	0.1359	Yes

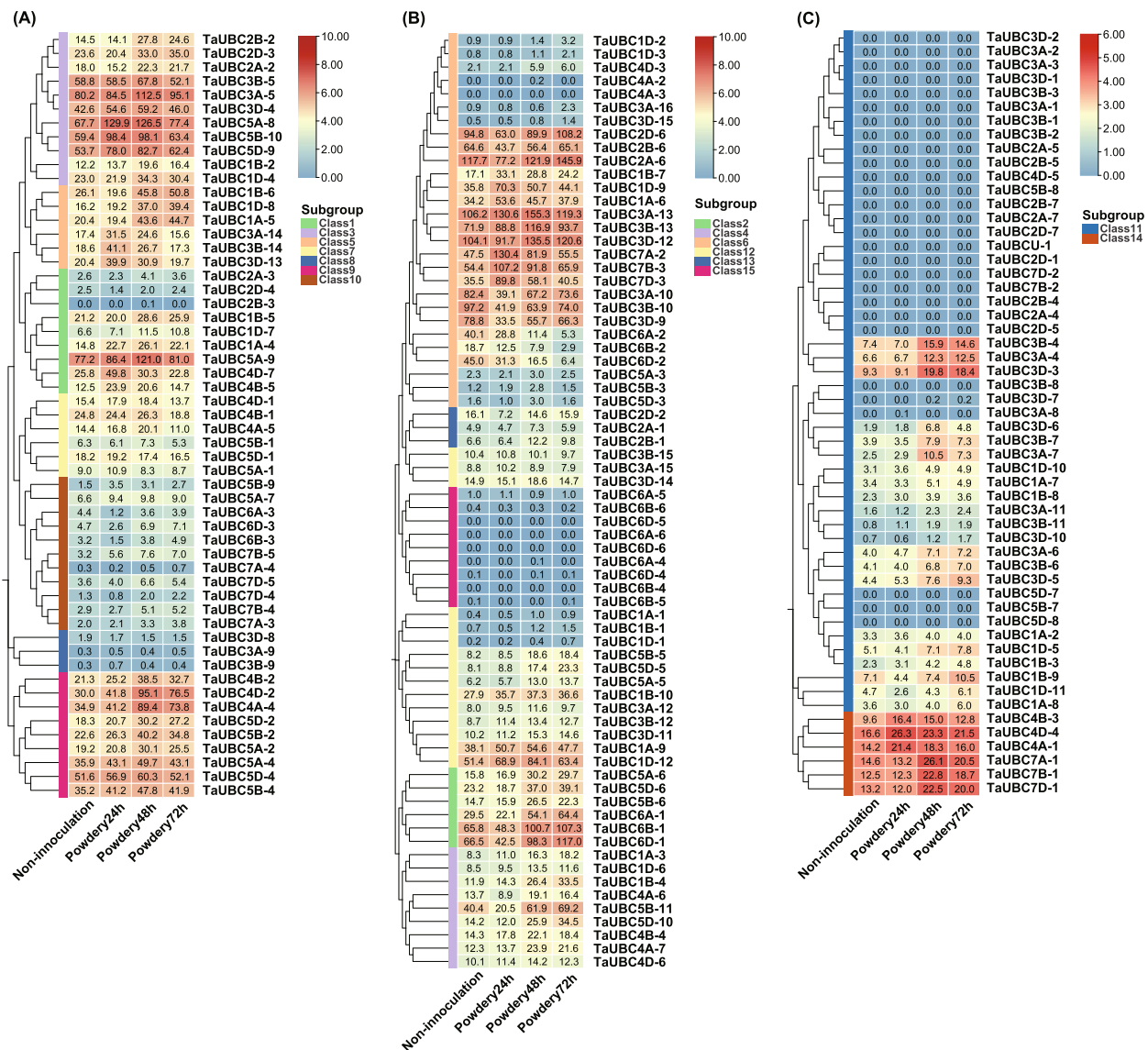


Fig. 4 Heat map of the expression profiling of *TaUBC* genes under different stresses. The color scale bar represents the expression values (in log₂-based tags per million values) of the genes, and the values in square frames represent the tags per million values. **A:** Members of Classes 1, 3, 5, 7, 8, 9 and 10. **B:** Members of Classes 2, 4, 6, 12, 13 and 15. **C:** Members of Classes 11 and 14

to *Bgt* in AK58, whereas silencing *TaUBC3B-14* enhances resistance to *Bgt* in AK58.

Protein active sites of *TaUBC3A-5* and *TaUBC3B-14*

The tertiary structure and active sites of the *TaUBC3A-5* and *TaUBC3B-14* proteins are shown in Fig. 8 with more detail in Table S4. Both proteins share a similar overall tertiary structure, consisting of five α -helices and four β -strands. Notably, the α 3, α 4, and α 5 helices in both proteins form an inverted cone-like structure. However, *TaUBC3A-5* contains two α -helices at its N-terminus and three at its C-terminus, while *TaUBC3B-14* features one α -helix at its N-terminus and four at its C-terminus. The active sites of *TaUBC3A-5* are primarily concentrated in the

N-terminal α 1 and α 2 helices and their associated pocket regions (Fig. 8A), whereas the active sites of *TaUBC3B-14* are mainly in the α 3, α 4, and α 5 helices and their corresponding pocket regions (Fig. 8B). Phylogenetically, *TaUBC3A-5* belongs to the Class 3, while *TaUBC3B-14* is part of the Class 5. The differences in protein structures and localizations of their active sites may underlie the distinct functional roles of *TaUBC3A-5* and *TaUBC3B-14* in wheat's response to powdery mildew infection.

Discussion

The E2/UBC ubiquitin-conjugating enzyme is a key component of the ubiquitin–proteasome system, and plays a vital role in numerous biological processes,

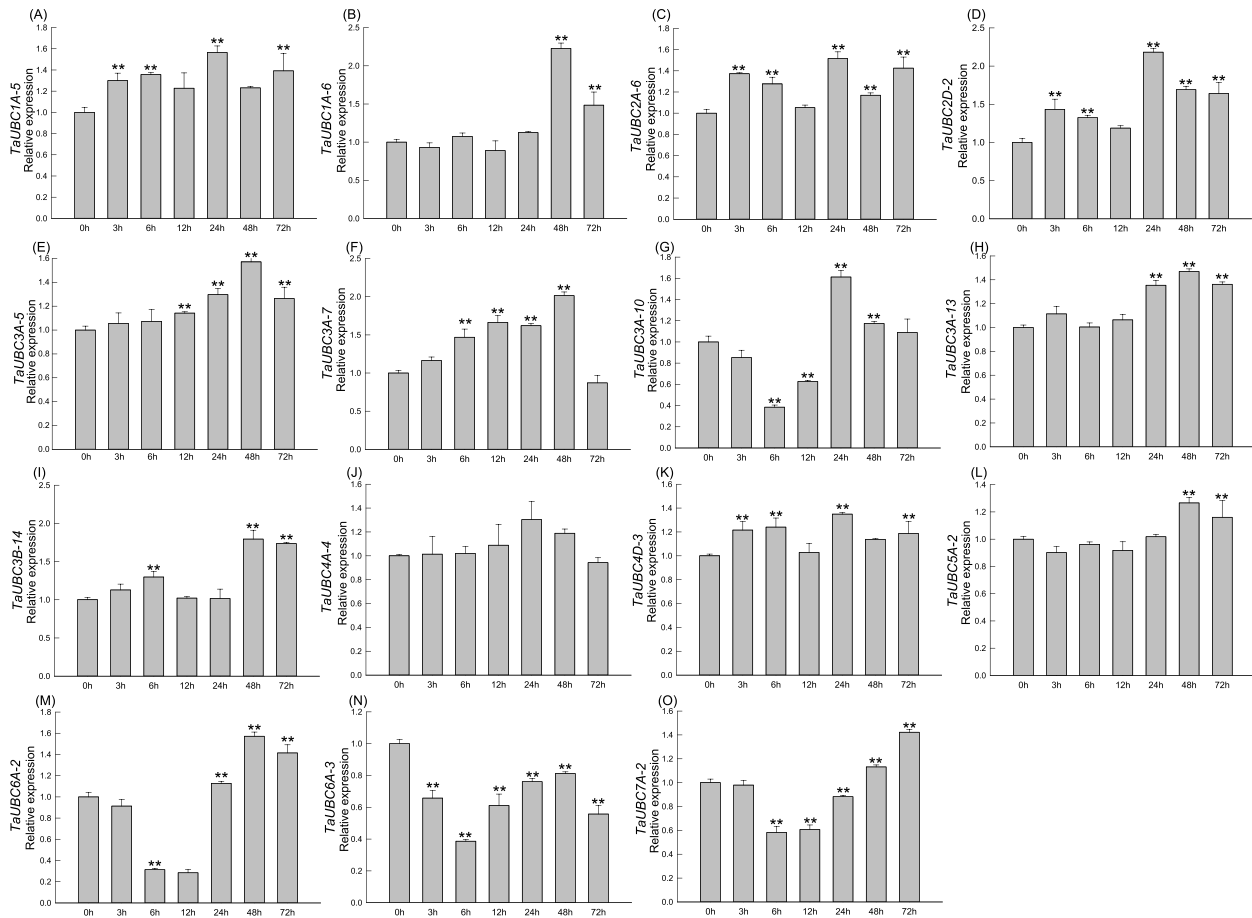


Fig. 5 Expression analysis of *TaUBCs* under *F. graminearum* by qRT-PCR. Asterisks indicate significant differences (assessed using Duncan's honestly significant difference test). ** $P < 0.01$. The values are the means of three technical replicates from one biological experiment. The raw data for qRT-PCR are listed in Table S2

including plant growth, development, and immune regulation [58]. However, its function in wheat powdery mildew and *F. graminearum* resistance has not been reported previously. In this study, 180 members of the wheat *UBC* gene family and 48 members of *Arabidopsis* were identified by sequence search in reference genome. Previous studies reported 37 *UBC* genes in *Arabidopsis* [23], and 12 additional *UBC* genes were identified in this study, which is likely due to differences in search methods used. Previous reports showed that 48 *UBC* genes were identified in rice [1], 75 in maize [22], 200 in *Brassica napus* [58], 57 in *Solanum tuberosum* [30], 40 in *Dimocarpus longan* [21], and 43 in *Vitis vinifera* [11], 74 in banana [10], and 83 and 84 were identified in *Pyrus bretschneideri* and *Pyrus communis*, respectively [5]. This study classified the identified *UBC* gene family members of wheat and *Arabidopsis* into 15 Classes, aligning with the classification of rice *UBC* genes [1], but exceeding the Classes reported for *Arabidopsis* [23]. Notably, each class contained members from both wheat and *Arabidopsis*, differing from previous findings

where some *Arabidopsis UBC* genes were listed separately and not grouped into classes [23], suggesting that *UBC* gene differentiation occurred prior to the divergence between monocotyledon and dicotyledon.

Gene duplication events are autonomous processes that helps plants to adapt to changing environments. Among the 180 members of the wheat *UBC* gene family, seven tandemly duplicated genes and eight segmentally duplicated genes were identified. This suggests that although most of the *UBC* genes studied originated from their ancestral species, gene duplication events were also crucial for the expansion and functional diversification of these gene families. The findings are consistent with the studies on other species, which demonstrated the importance of gene duplication in the expansion of the *UBC* gene families in *Pyrus bretschneideri* and *Pyrus communis* [5] as well as grape [11]. Additionally, analysis of conserved motifs and gene structures revealed that members of the wheat *UBC* gene families within the same Classes were more similar, while those from different classes

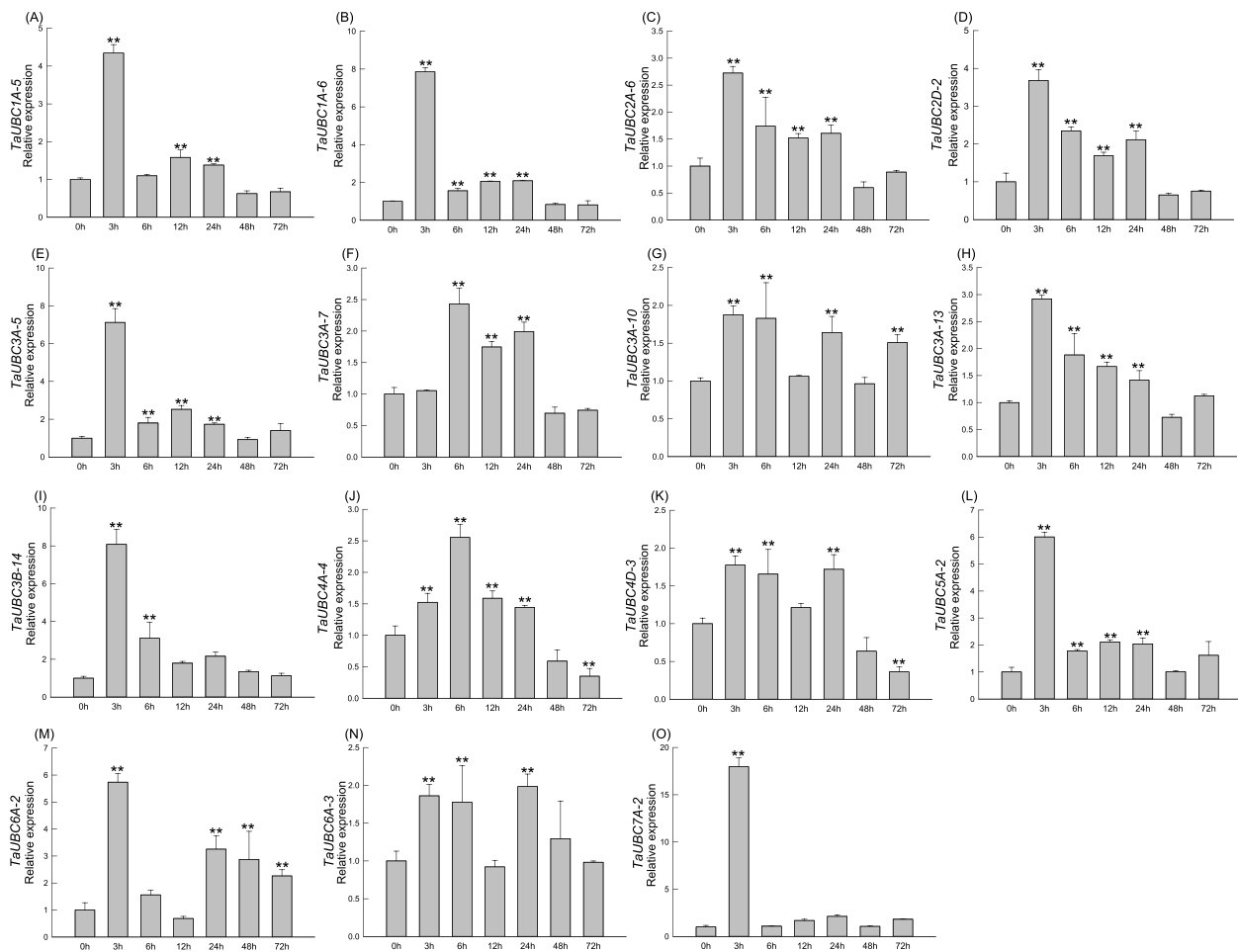


Fig. 6 Expression analysis of *TaUBCs* under *Blumeria graminis* f. sp. *tritici* stress by qRT-PCR. Asterisks indicate significant differences (assessed using Duncan's honestly significant difference test). ** $P < 0.01$. The values are the means of three technical replicates from one biological experiment. The raw data for qRT-PCR are listed in Table S2

exhibited distinct conserved motifs and gene structures. This indicates that the structural complexity of the *UBC* gene families contribute to its functional diversity.

The plant innate immune response is closely associated with ubiquitination-mediated protein degradation [52]. In *Arabidopsis*, inactivation of the *AtUBC13* gene impairs the protein-mediated immune responses involving nucleotide-binding domains and leucine-rich repeat sequences, leading to rapid proliferation of pathogens. This demonstrates the critical role of *AtUBC13* in the effector-triggered immunity process [47]. Furthermore, overexpression of *AtUBC35* enhances ubiquitination-mediated immune responses [44]. In this study, analysis of transcriptomic data from *Fusarium* stressed wheat plants revealed significant changes in the expression levels of some *TaUBC* genes. qRT-PCR results confirmed an increasing trend in the expression levels of some wheat *UBC* genes, such as *TaUBC1A-5* and *TaUBC1A-6*, they showed relatively modest increases, ranging from 1- 2.5

folds, but the increases were statistically significant. By comparison, the transcriptomic data powdery mildew infected plants showed significantly larger changes in the expression levels of some wheat *UBC* genes. qRT-PCR confirmed the substantial increase in the expression levels of *TaUBC3A-5* and *TaUBC3B-14*, ranging from 5–18 folds. VIGS experiments revealed that *TaUBC3A-5* (Class 3) silenced wheat plants increased susceptibility to *Bgt*, whereas *TaUBC3B-14* (Class 5) silenced wheat plants increased the resistance to *Bgt*. These findings suggest that different classes of wheat *UBC* genes play distinct roles in responding to attack from different pathogens. Gene redundancy may buffer organisms from loss-of-function mutations [6]. It is unclear whether *UBC* genes have functional redundancy. Other genes with highly similar sequences within the same Class may be silenced by VIGS, which limits the use of VIGS to verify the function of the target genes. Using gene editing to knock out the target genes is a good way to explore the functional redundancy of the *UBC* gene family members.

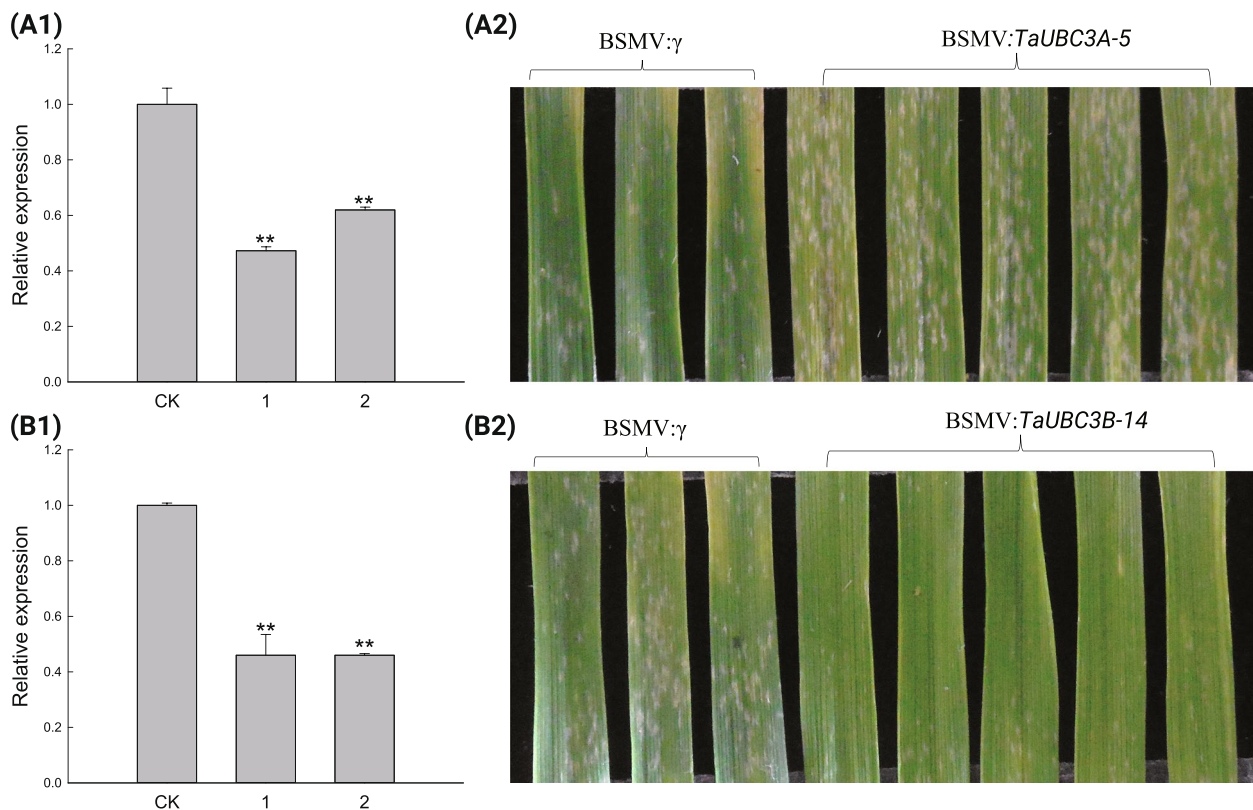


Fig. 7 Functional analysis of *TaUBC3A-5* and *TaUBC3B-14* by BSMV-VIGS in BainongAK58. Expression of *TaUBC3A-5* (A) and *TaUBC3B-14* (B) in the corresponding virus- infected leaves were checked by qRT-PCR. ** $P < 0.01$. BSMV: *TaUBC3A-5* (C) and *TaUBC3B-14* (D) infected plants were infected by *Bgt*, and photos were taken after six days of inoculation. Three biological replicates were used in the BSMV-VIGS experiment

Based on the previous research, *UBC13* is the most studied *UBC* gene. Numerous studies have shown that *UBC13* is essential for mammalian immune systems and *UBU13* interacts with an E3 enzyme TRAF6 to catalyze K63 polyubiquitination [20]. In plants, *AtUBC13* interacts with an F-box protein CPR1 that regulates the homeostasis of SNC1 to regulate the response to pathogen attacks and involved in programmed cell death [47]. The tomato *UBC13* protein Fni3 regulates disease resistance by interacting with the kinase protein Fen, plays a role in effector triggered immunity, and is required for Fen-mediated cell death [37]. The *OsUBC13* directly interacts with *OsSnRK1a*, the catalytic subunit of *SnRK1*. *OsUBC13* enhances the activity of *OsSnRK1a* to negatively regulate rice resistance to blast and bacterial leaf blight diseases, and the *OsUbc13*-RNAi lines showed lesion mimic phenotypes [29]. This suggesting that *UBC* genes may relate to the reactive oxygen species signaling pathway. Although there have been some reports on the function of *UBC13* in plant immunity, it is still unclear which E3 ligases bind to it. Current studies showed that the proteins directly bound to *UBC13* in different species were different. To our knowledge, few E3 were reported to interact with *UBC13*. In this study, *TaUBC3B-14* is an ortholog of *AtUBC13* in wheat (Fig. 1), and *TaUBC3B-14*

negatively regulates powdery mildew disease resistance. However, further studies remain needed to uncover the resistance mechanism and the substrates or E3 ligases for *TaUBC3B-14*.

A previous study has shown that silencing ubiquitin-binding enzyme *TaU4* delays the onset of wheat smut (*Tilletia foetida*) and Septoria leaf blotch, indicating it's a negative regulator in defending against those pathogens [34]. Comparative analysis revealed that *TaUBC3A-5* and *TaU4* are the same gene, suggesting that *TaUBC3A-5*/*TaU4* may exhibit functional versatility to address different wheat biotic stresses. Structural differences among homologous genes are known to drive functional divergence, though the underlying mechanisms remain unclear [54, 56]. Active site analysis showed that *TaUBC3A-5* (Class 3) and *TaUBC3B-14* (Class 5) differ in their three-dimensional structures and active sites. The active sites of functional proteins are generally on the alpha helix or the structure formed by multiple active sites, and the positions of the active sites should be easy to bind. Based on the analysis of three-dimensional structure, the active sites of *TaUBC3A-5* and *TaUBC3B-14* are relatively easy to contact with other proteins. However, the active sites of *TaUBC3A-5* are mainly concentrated on the N-terminal alpha helix, while those of

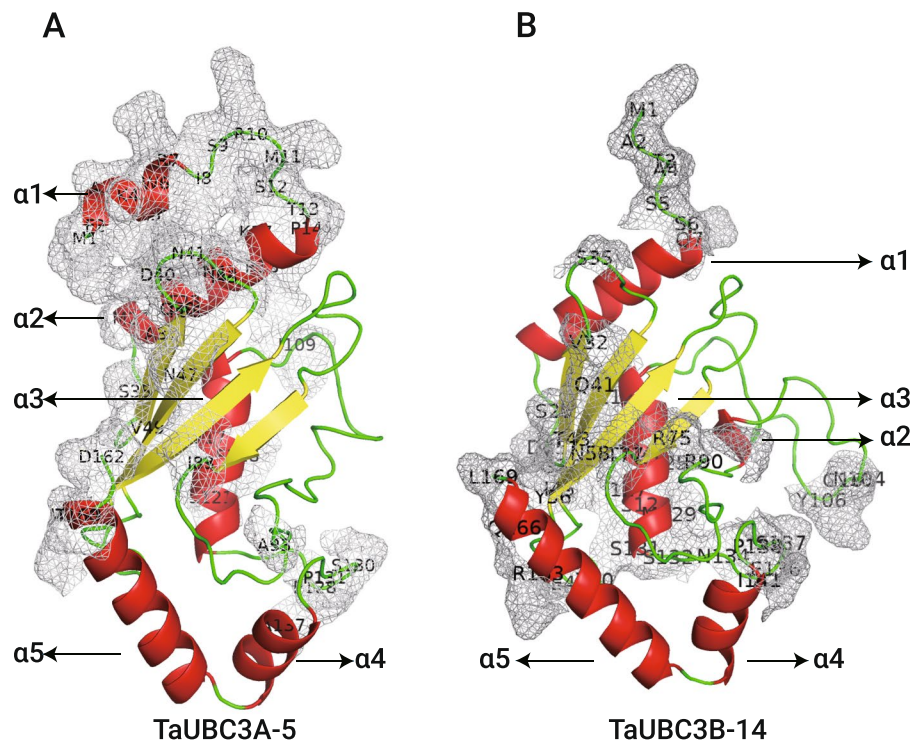


Fig. 8 The protein structure and active sites of TaUBC3A-5 and TaUBC3B-14. All active sites are listed in Table S4, Mesh coverage and amino acid labeling represent protein active sites. **A:** The three-dimensional structure and active sites of TaUBC3A-5, **B:** The three-dimensional structure and active sites of TaUBC3B-14

TaUBC3B-14 are mainly concentrated on the C-terminal alpha helix (Fig. 8). This may imply that they may bind to different proteins. We hypothesize that these structural variations may contribute to their functional differentiation in responding to wheat powdery mildew infection.

In conclusion, this study utilized a combination of bioinformatics and molecular biology approaches to comprehensively identify and analyze members of the wheat *UBC* gene family at the whole-genome level. As a result, 180 *UBC* genes were identified in wheat and categorized into 15 Classes. Through analysis of *UBC* gene expression pattern under wheat powdery mildew stress and functional validation via VIGS, we found that *TaUBC3A-5* and *TaUBC3B-14* may have opposite roles in the responses to powdery mildew stress. These findings provide valuable target genes for further research on wheat *UBC* gene family in powdery mildew disease resistance and for genetic improvement of wheat resistance to multiple disease.

Abbreviations	
UBC	Ubiquitin-conjugating enzyme
FHB	<i>Fusarium</i> Head blight
<i>Bgt</i>	<i>Blumeria graminis</i> F. sp. <i>tritici</i>
AK58	Bainong AK58
HMM	Hidden Markov Model
MCSanX	Multiple Collinearity Scan toolkit
VIGS	Virus-induced gene silencing

Supplementary Information
The online version contains supplementary material available at <https://doi.org/10.1186/s12870-025-06931-x>.

- Supplementary Material 1.
- Supplementary Material 2.
- Supplementary Material 3.
- Supplementary Material 4.
- Supplementary Material 5.
- Supplementary Material 6.
- Supplementary Material 7.

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Authors' contributions
Ping Hu, Lu Zhang and Jun Xu completed the experimental design, experimental process, data analysis, initial draft of the paper, and writing revisions for this study; Lu Zhang, Yueming Ren, Mengfei Wang, Wanglong Luo, Kaige Zhou, Xinjie Zhu and Zhiyuan Liu completed the experimental process and experimental data; Chengwei Li and Haiyan Hu participated in experimental design and paper writing revision. All authors have read and agreed to the final text.

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Data availability

Datasets supporting the findings of this study are available within the article and its supplementary materials.

Declarations

Ethics approval and consent to participate

Not applicable.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

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